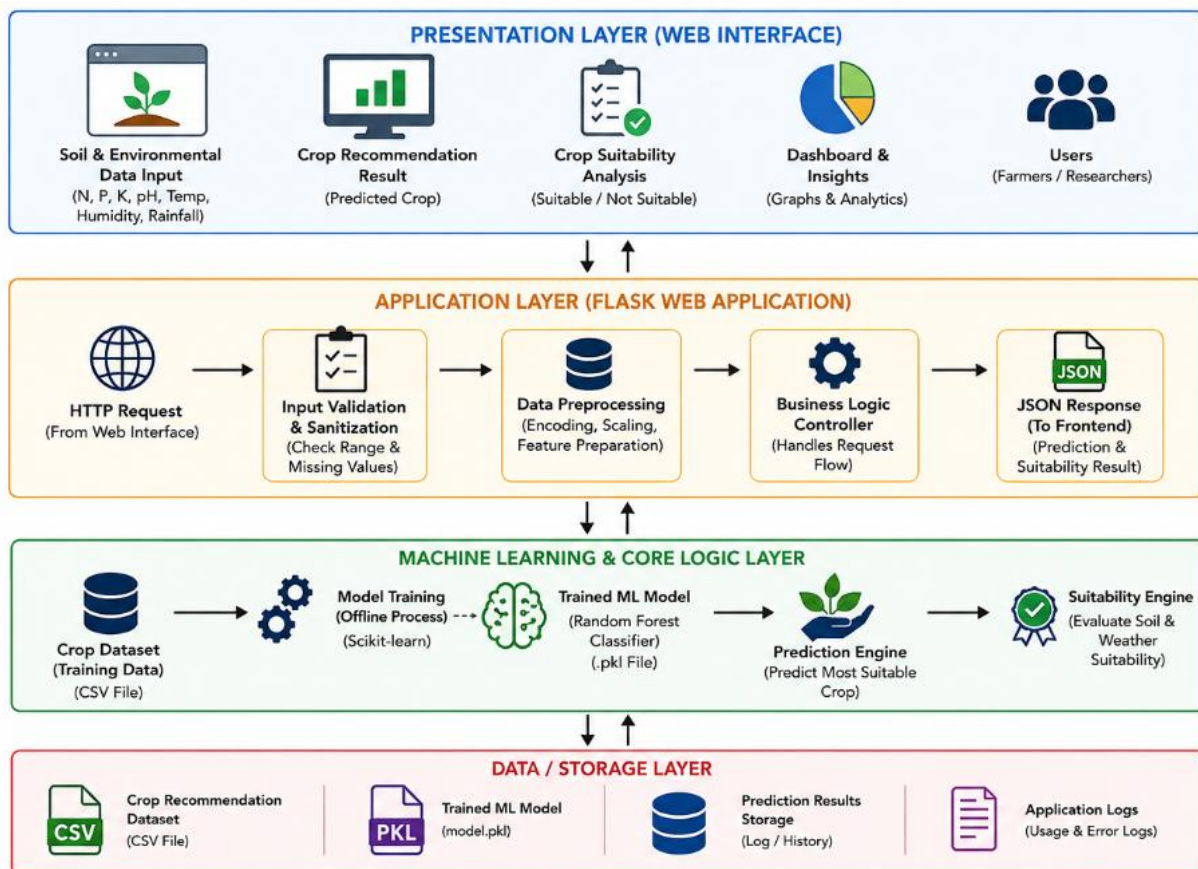


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a simple web interface where users enter soil and environmental parameters and view crop recommendations, suitability analysis, and prediction results.	HTML5, CSS3, JavaScript
API Gateway / Request Handler	Receives user requests from the web application, validates the input, forwards data to the machine learning module, and returns prediction results to the user.	Flask, Python
Core Logic / ML	Processes soil data, performs data preprocessing, executes the trained machine learning model, and generates crop recommendations and suitability analysis.	Python, Scikit-learn, Pandas, NumPy
Data / Storage Layer	Stores the crop recommendation dataset, trained machine learning model, and prediction-related data required by the application.	CSV Dataset, Pickle (.pkl), Local File Storage